

RANJAN KUMAR

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RESEARCH SNAPSHOT

India's agriculture is highly dependent on monsoon rainfall, yet the underlying drivers of rainfall variability remain inadequately understood. In particular, the relative contributions of local versus remote moisture sources and the governing atmospheric processes are still not clearly quantified.

My research addresses this gap by applying Lagrangian moisture tracking approaches to quantify atmospheric moisture transport and its contribution to precipitation across the Indian subcontinent. The work further decomposes the roles of dynamic and thermodynamic processes in shaping rainfall variability.

This enables a process-based understanding of how different moisture sources influence precipitation across diverse agro-climatic regions of India, with implications for improving climate resilience, water resource planning, and agricultural sustainability.

KEY RESEARCH CONTRIBUTIONS

- Most rainfall over India is driven by large-scale atmospheric winds carrying moisture from oceans, while short-term weather disturbances further intensify rainfall variability, especially over central and northern India.
- Demonstrated that oceanic moisture sources (Western Indian Ocean, Arabian Sea and Bay of Bengal) contribute significantly more to rainfall than local land-based evaporation (Ganga basin).
- Showed that steady background circulation controls overall rainfall patterns, whereas transient weather systems are responsible for extreme rainfall events.

RESEARCH INTERESTS

Atmospheric Moisture Transport, Land–Atmosphere Interactions, Hydroclimatology, Monsoon Dynamics and Variability, Moisture Budget Diagnostics, Extreme Precipitation and Atmospheric Rivers, Climate Variability and Teleconnections, Agro-Climatic Systems

PUBLICATIONS

- Kumar, R., Pathak, A., & Kulkarni, S. (2026). Climatic and transient controls on Indian monsoon E–P variability using moisture-budget decomposition. *Environmental Research Communications*.(Q1)
- Kumar, R., & Pathak, A. (2025). Lagrangian quantification of atmospheric moisture sources for extreme rainfall events over India during 2023 summer monsoon. *Science of the Total Environment*.(Q1)
- Kumar, R., Ghosh, S., & Singh, L. K. (2025). Mapping Wheat Potential: A GIS-Based Agro-Land Suitability Analysis. *Frontiers in Agronomy*.(Q1)

- Kumar, R., Joseph, A., & Poltharana, A. (2026). Experimental Investigation on Sustainable Soil Stabilization Using Coconut Coir Fibre. *Journal of Scientific Research and Reports*.
- Kumar, R., & Patel, G. R. (2020). Assessment of Agro-Land Suitability for Rice using GIS and Remote Sensing. *International Journal of Current Microbiology and Applied Sciences*.

CONFERENCES

- Kumar, R., & Pathak, A. (2022, December). Diagnostic Evaluation of Atmospheric Moisture Budget (EP) during ENSO years and its relation to ISM rainfall variability. *AGU Fall Meeting Abstracts*, Vol. 2022, H44C-07.
- Kumar, R., Pramod, & Pathak, A. (2024, June). Detection of atmospheric rivers associated with extreme precipitation over the Indian subcontinent. *International Atmospheric Rivers Conference 2024*, Scripps Institution of Oceanography, La Jolla, USA. Book of Abstracts.
- Kumar, R., Chouhan, P., & Pathak, A. (2024, December). Identification of Atmospheric Rivers Linked to Extreme Precipitation in the Core Monsoon Zone of India. *AGU Fall Meeting Abstracts*, Vol. 2024, A07-03.
- Kumar, R., & Pathak, A. (2024, December). Role of Western Indian Ocean on the Indian Monsoon Extreme Events during 2023. *AGU Fall Meeting Abstracts*, Vol. 2024, A43Y-03.
- Kumar, R., & Pathak, A. (2025, April). Tracing Moisture Flow from the Bay of Bengal and Arabian Sea with its Impact on Ganga Basin during Monsoonal Extremes. *EGU General Assembly Conference Abstracts*, EGU25-4531.
- Kumar, R., & Pathak, A. (2025, October 6). *Atmospheric moisture budget and its variability over the Indian subcontinent*. XIIth Scientific Assembly of the International Association of Hydrological Sciences (IAHS 2025), IIT Roorkee, India.
- Kumar, R., & Pathak, A. (2026). Chasing the sources of India's extreme rainfall: Insights from UTrack and HYSPLIT. Presented at Roorkee Water Conclave 2025–26, IIT Roorkee, India.

SKILLS

Programming Languages: Python, MATLAB, R

Scientific / Domain Languages: NCL (NCAR Command Language)

Python Libraries: NumPy, Pandas, Matplotlib, Xarray

Data Formats & Processing: NetCDF, CDO, NCO

Geospatial Software: ArcGIS, QGIS, ERDAS, Google Earth Engine

Computing & Tools: Linux, Bash, GitHub, Remote HPC

Models / Frameworks: UTrack Moisture Tracking Model, WRF Model

EDUCATION

- **Ph.D., Agricultural Engineering** 2021–Present
 IIT Kharagpur (Expected completion: 2026) CGPA: 8.77/10
Advisor: Prof. Amey Pathak

- **M.Tech, Agricultural Engineering** 2019
CAET, Anand Agricultural University (AAU) CGPA: 8.78/10
- **B.Tech, Agricultural Engineering** 2017
KCAET, Kerala Agricultural University (KAU) CGPA: 7.70/10
- **AISSCE (12th)** 2012
CBSE 69.6%
- **AISSE (10th)** 2010
CBSE 80%

RESEARCH EXPERIENCE / INTERNSHIPS

- Served as a Teaching Assistant for Engineering Drawing (CE13001) at IIT Kharagpur during 2020–21 and 2021–22.
- Served as a Teaching Assistant in Agricultural and Food Engineering at IIT Kharagpur for Irrigation and Drainage Engineering Lab and Geo-informatics Lab during 2021–23.
- Completed internship at Indian Institute of Remote Sensing (IIRS), Dehradun under NRMS (Dept. of Space, Govt. of India) from August to November 2015.

AWARDS & ACHIEVEMENTS

• Academic Achievements:

- Achieved AIR-901 in ICAR-AIEEA (UG), 2013.
- Secured Rank 352 in Bihar Combined Entrance Competitive Examination (BCECE), 2013.
- Secured Rank 681 in Uttar Pradesh Combined Entrance Test (UPCET), 2013.
- Qualified JEE Mains and JEE Advanced, 2013.
- Achieved AIR-42 in ICAR-AIEEA (PG), 2017.
- Achieved AIR-36 in ICAR AICE-JRF/SRF (Ph.D.), 2019.
- College Topper in M.Tech Gold Medalist (2017–2019), Anand Agricultural University.
- Secured Rank 3 in IIT Ph.D. entrance exam (2019) among selected candidates.

• Awards & Recognition:

- Awarded memento certificate for volunteering under MSME program, IIT Kharagpur (2024).
- Awarded Pt. Ishwar Chandra Vidyasagar Award (2024) for contributions to student welfare at IIT Kharagpur.

• Sports & Leadership:

- Represented Kerala Agricultural University in All India South Zone Cricket Tournament, Pondicherry (2013).
- Participated in All India Inter-Agricultural University Games (Badminton), Coimbatore (2016).
- Participated in South Zone Badminton Tournament, TNAU Coimbatore (2016).
- Gold Medal – Intercollegiate Badminton and Table Tennis Championship, AAU (2018).
- Gold Medal – Interhall Badminton Tournament, IIT Kharagpur (2022, 2023).
- Silver Medal – Interhall Badminton Tournament, IIT Kharagpur (2025).
- Served as General Secretary, Vidyasagar Hall of Residence, IIT Kharagpur (2022–2024).

• Other Activities:

- Volunteered in National Service Scheme (NSS), 2016.

REFERENCES

Dr. Amey Pathak

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